

To Enhance Municipal Revenue Department Processes Using Process Mining Techniques (A Case Study: Hamedan Province)

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Abstract—Over the past decade, process mining has emerged as a research field analyzing processes using event data. Unlike traditional data mining techniques, which focus on specific steps, process mining looks at end-to-end business processes. This is made possible by the increasing availability of event data and advancements in process discovery and conformance checking techniques. Process models, which were previously created manually without using event data, are now used for analysis, simulation, verification, and enactment by Business Process Management and Workflow Management systems (BPM/WFM). Process mining techniques utilize event logs to discover, analyze, and enhance business processes by examining the activities performed by people, machines, and software. In this paper, four process mining miners (Alpha Miner, Alpha++ Miner, Genetic Miner and Heuristics Miner) were utilized to enhance the processes of revenue department in the municipality of Hamadan based on the event logs. The contextualization of this research lies in its importance in improving municipal revenue practices, enhancing transparency, and optimizing resource allocation, which is a critical aspect of efficient governance. Process mining offers a data-driven approach to identify bottlenecks, delays, and deviations in the revenue process, ultimately leading to more effective decision-making and improved public service delivery. The results demonstrated that the Genetic Miner achieved the best result with an impressive 98% overall score, which is an average across key performance metrics including fitness, simplicity, precision, and generalization. The implications of this finding highlight the potential for municipalities and similar organizations to leverage process mining as a means of enhancing their revenue processes, thereby fostering more efficient resource allocation and governance.

Keywords—Process Mining, Revenue Process Enhancement, Hamedan Province, Hamedan Municipality, Event Log

I. INTRODUCTION

Process mining is a powerful approach aiming to discover, monitor, and improve real processes by analyzing event logs available in information systems. Over the past decade, process mining techniques have significantly matured, and the abundance of event data has grown exponentially. This has allowed management trends related to process improvement and compliance to benefit from process mining.

The process mining journey begins with an event log, where each event represents a specific activity within a process and is associated with a particular case or process instance. Events within a case are ordered, forming a single "run" of the process. Event logs may contain additional information, such as the resource executing the activity, timestamps, or data elements associated with the event.

Process mining encompasses three types of analysis. The first type is process discovery, which automatically generates a process model solely based on the event log, without prior knowledge. Many organizations find it surprising that existing techniques can accurately uncover real processes using historical behaviors stored in event logs. The second type is conformance checking, where an existing process model is compared with an event log to assess the level of alignment between the model and the recorded reality. The third type is process enhancement, which utilizes the information captured in the event log to extend or improve an existing process model. [1]

Process mining differs from traditional Business Process Management (BPM) techniques that rely on manually created models. It is characterized by being process-centric, intelligent (learning from historical data), and fact-based (based on event data rather than subjective opinions). Process mining enables evidence-based BPM by extracting knowledge from observed behaviors recorded in event logs. [2]

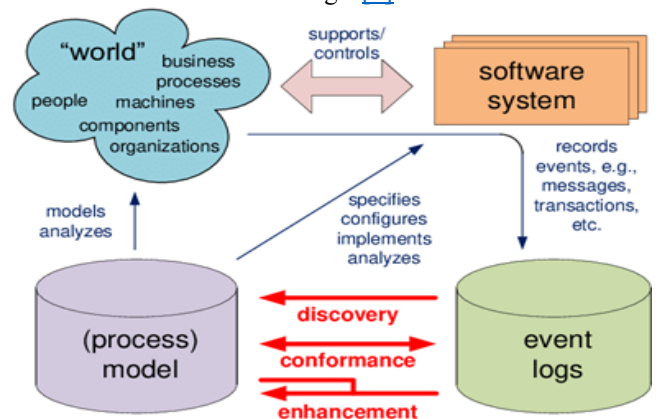


Fig. 1. The three basic types of process mining: (a) Discovery, (b) Conformance, and (c) Enhancement [2]

Process mining is closely related to data mining, although it differs in its focus. While classical data mining techniques are primarily data-centric, process mining takes a process-centric approach. Mainstream business process modeling techniques, such as the Business Process Modeling Notation (BPMN), UML activity diagrams, Event-driven Process Chains (EPC), and Petri nets, are commonly used in process mining.

These notations enable the modeling of processes with concurrency, choice, iteration, and other essential characteristics. In contrast to data mining, which primarily analyzes datasets, process mining analyzes event logs and focuses on understanding and improving the underlying processes.

A. Event Logs

Digital event data is pervasive across all sectors, economies, organizations, and households, and its exponential growth is expected to continue. This abundance of data presents new opportunities for process analysis, where insights can be derived from observed facts rather than relying solely on manually created models. The key starting point for process mining is an event log, which captures the sequence of events and serves as the foundation for analyzing and improving processes.

For this experiment, event logs were obtained from the municipality of Hamadan province. These event logs were subsequently refined, simplified, and structured into tabular formats for the purpose of analysis and improvement. The primary objective of this paper is to enhance the revenue department processes within the municipal authority of Hamedan province. Our goal is to extract a comprehensive process model that vividly illustrates the execution and allocation of resources within this process. To achieve this, we employ four process mining techniques, including the Alpha Miner, Alpha++ Miner, Genetic Miner, and Heuristics Miner.

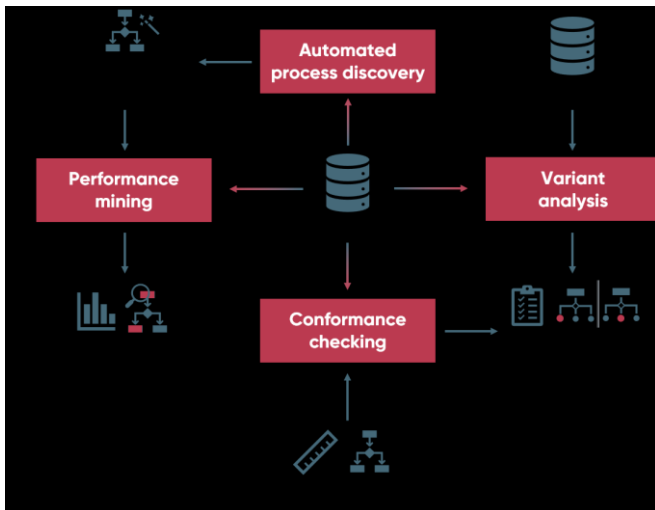


Fig. 2. Process mining and event logs [3]

II. RELATED WORK

Paper [4] aims to investigate students' behavior and interaction patterns in online quiz-based activities within learning management systems (LMS). The study utilizes a process-oriented approach and process mining methods to identify specific types of interaction sequences, providing insights into students' quiz-taking strategies in LMS, with

implications for both researchers and practitioners in online education.

Paper [5] seeks to assess the usability of Process Mining approaches in the context of Smart City applications by addressing three key research questions: 1) Identifying the most researched Smart City issues using Process Mining; 2) Identifying the prevalent Process Mining methods employed in Smart City applications; 3) Identifying Smart City areas where Process Mining has not been extensively applied. The study aims to uncover challenges and opportunities in applying cutting-edge Process Mining techniques to address emerging problems in Smart Cities.

Paper [6] explores the use of process mining in self-regulated learning (SRL) research, specifically focusing on analyzing temporal and sequential aspects of SRL processes. It evaluates four common process mining algorithms (Inductive Miner, Heuristics Miner, Fuzzy Miner, and pMineR) in an undergraduate education context. The study finds that Fuzzy Miner and pMineR provide valuable insights for SRL research, and combining their metrics enhances the understanding of temporal and sequential relationships in SRL processes. The paper suggests future SRL research should consider using a mix of process mining algorithms and explore dedicated tools for SRL analysis.

Paper [7] focuses on improving the assessment of in-hospital mortality risk for diabetes ICU patients, who face elevated complications and risks during their stay. To address this challenge, the paper introduces a novel approach integrating process mining and deep learning techniques. It leverages patients' medical history by converting past hospital encounter records into event logs suitable for process mining. Subsequently, a process model is constructed to depict the patients' historical hospital experiences. To predict in-hospital mortality, this process model is combined with demographic and medical information, offering enhanced performance compared to traditional severity scoring methods and standard machine learning approaches. The paper substantiates these improvements through experimentation with the Medical Information Mart for Intensive Care III dataset.

III. MATERIALS AND METHODS

In the sections that follow, first, event log selection will be discussed, then, each of the mining materials and methods will be comprehensively introduced. Subsequently, the focus will shift to the evaluation phase, where a detailed exploration of the various performance indicators and metrics used to assess the quality and efficacy of the process models generated through the examination of event data will be undertaken. This thorough examination of performance evaluation serves as a critical cornerstone in our research, enabling informed and data-driven decisions to be made regarding the processes and their potential for optimization and enhancement. By these metrics, valuable insights can be gained into which mining technique yields the most accurate, efficient, and effective results, thereby guiding the municipality toward a more streamlined and productive revenue process.

A. Event Log Selection

For the process enhancement, the municipal revenue department processes were asked in 2023. Event logs were collected based on these processes, which were then summarized into tables. The Processes had 10 steps and the event logs had 29240 instances which were made during the processes. Subsequently, these event logs were imported into “ProM 6.12” software via XML files and utilized in four distinct miners: the Alpha Miner, Alpha++ Miner, Genetic Miner, and Heuristics Miner.

B. Process Mining Miners

1) Alpha Miner (AM)

The Alpha Miner is a process discovery algorithm used in process mining. It is one of the most basic and widely used process mining techniques for discovering a process model from event logs. The Alpha Miner algorithm works by identifying the most common causal relationships between activities in the event logs. It starts by identifying the activities that frequently occur as the first and last activities in the log. Then, it iteratively explores the event logs to discover the direct succession relations between activities. This process continues until all possible direct causal dependencies have been identified. The resulting process model is represented as a Petri net or a process flow diagram, which shows the causal relationships between activities in the process. However, it's important to note that the Alpha Miner may produce overly complex models or fail to capture certain process behaviors in more complex scenarios. [8]

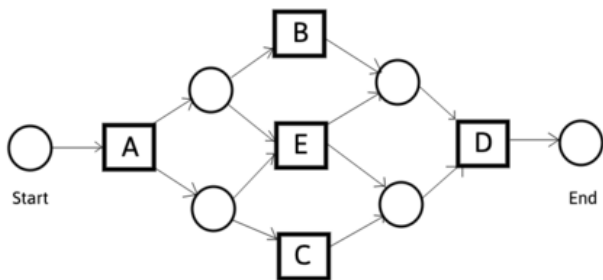


Fig. 3. Alpha algorithm [9]

2) Alpha++ Miner (AM++)

Alpha++ Miner is an extension of the Alpha Miner algorithm in process mining. It is an enhanced version addressing some limitations of the original Alpha Miner. The Alpha++ Miner algorithm aims to improve the accuracy and robustness of process discovery by incorporating additional techniques. One key enhancement of Alpha++ Miner is the consideration of explicit and implicit dependencies between activities. While the original Alpha Miner only focuses on direct causal dependencies, Alpha++ Miner also considers indirect dependencies between activities. This means that it can capture more complex process behaviors and dependencies that may not be immediately apparent in the event logs. Another improvement in Alpha++ Miner is the handling of noise and inconsistencies in the event logs. It employs techniques to filter out noise and handle missing or incomplete data, leading to more reliable process models. Furthermore, Alpha++ Miner incorporates various heuristics

and optimization strategies to refine the discovered process model, aiming to simplify the model structure and improve its interpretability. Overall, Alpha++ Miner extends the capabilities of the original Alpha Miner algorithm, allowing for more accurate and comprehensive process discovery in process mining. [10]

3) Genetic Miner (GM)

Genetic Miner is a process discovery algorithm used in process mining. It is inspired by genetic algorithms, which are search and optimization algorithms based on the principles of natural selection and evolution. The Genetic Miner algorithm applies these principles to discover process models from event logs. The Genetic Miner algorithm starts with an initial population of randomly generated process models. Each model in the population represents a potential solution. These models are evaluated based on their fitness, which measures how well they fit the event log data. The algorithm then applies genetic operators, such as selection, crossover, and mutation, to evolve the population. Through the selection process, models with higher fitness are more likely to be selected for reproduction. Crossover involves combining parts of two parent models to create new offspring models, while mutation introduces small random changes to the models. This process mimics the idea of genetic variation and inheritance in biological evolution. After each generation, the fitness of the models is re-evaluated, and the process of selection, crossover, and mutation is repeated. This iterative process continues until a termination condition is met, such as reaching a maximum number of generations or achieving a desired level of fitness. [11]

The final output of the Genetic Miner algorithm is a process model representing the discovered process based on the event log data. This model can be represented in various notations, such as Petri nets or process flow diagrams.

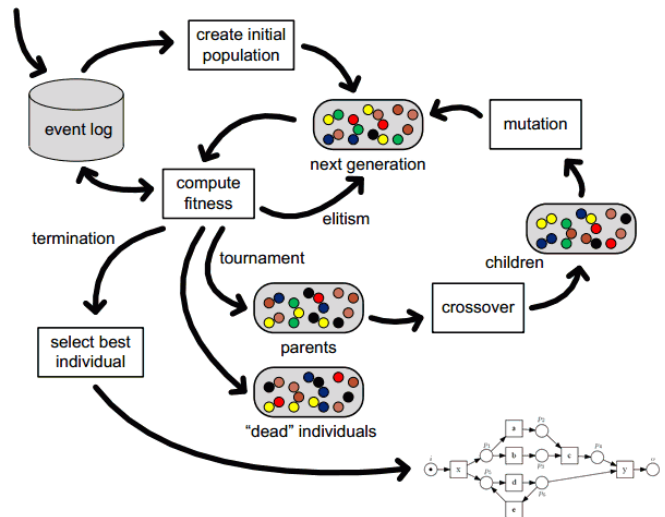


Fig. 4. Genetic miner [12]

4) Heuristics Miner (HM)

Heuristics Miner is a process discovery algorithm used in process mining. It is a popular and effective technique for automatically deriving process models from event logs. The Heuristics Miner algorithm focuses on capturing the order and dependencies between activities in the process. The Heuristics Miner algorithm works by analyzing the event logs to identify frequent behavioral patterns, or heuristics, that represent common process execution paths. These heuristics are then

used to construct a process model reflecting the observed behavior in the event logs. The algorithm starts by identifying the most frequent start and end activities in the event logs. It then examines the order in which activities are typically executed and constructs a direct succession relation between them. This relation indicates the most common transition between activities based on their co-occurrence in the log.

Additionally, the Heuristics Miner algorithm incorporates various heuristics to capture more complex dependencies between activities. For example, it considers the notion of concurrency, where activities can occur simultaneously. It also accounts for skipping and insertion behaviors, which indicate optional or extra activities that may occur in the process. The Heuristics Miner algorithm uses a combination of heuristics to construct a process model representing the observed behavior in the event log. This model can be represented as a Petri net, a process flow diagram, or another suitable notation. [13]

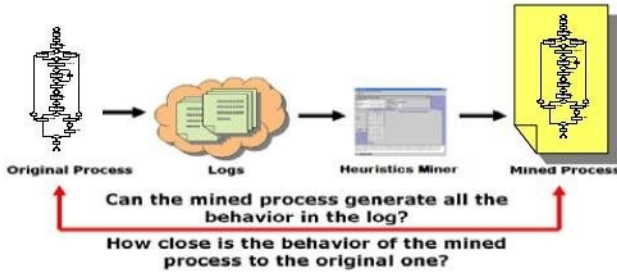


Fig. 5. Heuristics miner [14]

One advantage of Heuristics Miner is its ability to handle noisy and incomplete event logs. It is robust to variations and exceptions in the data, making it suitable for real-world process mining scenarios.

C. Quality Dimension

In the field of Process Mining, key metrics are employed to assess the quality and effectiveness of process models derived from the analysis of event data. These metrics, commonly known as performance indicators, include 'fitness', 'simplicity', 'precision', and 'generalization'. They play a fundamental role in quantifying the degree to which a process model accurately reflects actual process behavior and its suitability for process analysis and improvement. These performance indicators are essential for evaluating the quality and utility of process mining outcomes. [15]

1) Fitness

In process mining, fitness refers to the measure of how well a discovered process model matches the observed behavior in the event log. It assesses the degree of alignment between the process model and the recorded events. Fitness is typically calculated by comparing the executed behavior of the process model with the actual observed behavior in the event log. It quantifies the extent to which the model accurately reproduces the sequence and timing of activities as recorded in the log.

$$f = \frac{1}{2} \left(1 - \frac{\sum_{i=1}^k n_i m_i}{\sum_{i=1}^k n_i c_i} \right) + \frac{1}{2} \left(1 - \frac{\sum_{i=1}^k n_i r_i}{\sum_{i=1}^k n_i p_i} \right). \quad (1)$$

2) Simplicity

Simplicity refers to the measure of how easily understandable and straightforward a process model is. It assesses the complexity and clarity of the model representation. Simplicity is important because a simple process model is easier to interpret, communicate, and maintain. It allows stakeholders, such as process analysts, managers, or end-users, to comprehend the model and its underlying process easily. A simple model facilitates effective decision-making, process improvement, and communication among stakeholders.

$$\partial s = \frac{|T| - (|T_{DT}| + |T_{IT}|)}{T} \quad (2)$$

3) Precision

Precision refers to the accuracy and correctness of the discovered process model or the results obtained from analyzing event data. It measures how well the identified process model represents the actual behavior of the underlying process. Precision is crucial in process mining because the reliability of the insights and decisions made based on the process model depends on its accuracy. A high level of precision ensures that the discovered model faithfully captures the real-world process, enabling effective analysis, improvement, and decision-making.

$$\alpha_B = \frac{\sum_{i=1}^k n_i (|t_v| - x_i)}{(|t_v| - 1) \sum_{i=1}^k n_i} \quad (3)$$

4) Generalization

Generalization refers to the process of abstracting and simplifying the discovered process model or event data to reveal higher-level patterns, trends, or commonalities. It involves reducing the complexity and level of detail in order to extract more general insights about the process. Generalization plays a crucial role in process mining as it helps to uncover the underlying structure and characteristics of the process, making it easier to understand and analyze. By identifying common patterns and behaviors, generalization allows for a more concise and higher-level representation of the process, which can aid in decision-making and process improvement efforts.

$$Q_g = 1 - \frac{\sum_1^T (\sqrt{\# \text{ execution}})^{-1}}{T} \quad (4)$$

In conclusion, the balance of the four quality dimensions, namely fitness, simplicity, precision, and generalization, is paramount in the realm of process mining. Achieving this equilibrium is essential to ensuring that process models effectively serve their intended purposes and contribute to organizational success.

In practice, finding the optimal equilibrium among these four dimensions is a delicate and context-specific endeavor. The right balance will vary depending on the unique requirements, goals, and constraints of each process mining project. Therefore, a thoughtful approach to modeling and continuous monitoring and adjustment is crucial. Achieving this balance ensures that process models are not only faithful representations of real-world processes but also accessible,

precise, and adaptable, making them valuable tools for process analysis and improvement.

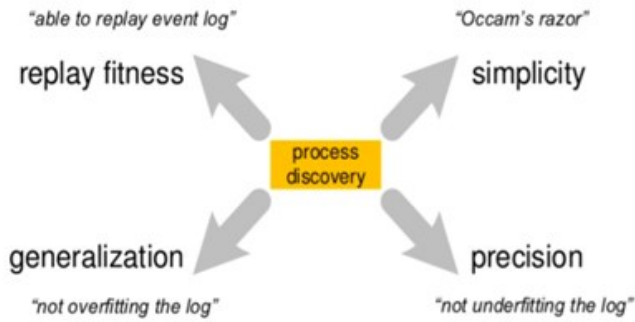


Fig. 6. The balance of four quality dimensions [15]

IV. RESULTS

Closely examined in this article are the results pertaining to the Quality Dimension and the accompanying models. Our evaluation encompasses critical metrics, including fitness, simplicity, precision, and generalization, to discern the degree of alignment between each model and the actual revenue processes within the municipality. Fitness gauges the accuracy of the model's representation of real-world processes, while simplicity measures its complexity. Precision evaluates the model's ability to predict and adhere to expected outcomes, and generalization assesses its adaptability to diverse scenarios. Furthermore, an overall score, derived from the aggregation of these metrics, aids in the selection of the most suitable model for enhancing the processes of revenue department at Hamedan province municipality.

A. QUALITY DIMENSION RESULTS

1) Fitness

In reference to Fig. 7., the Genetic Miner stands out with the highest fitness score, reaching an impressive value of 0.951. This particular metric, fitness, indicates the extent to which the Genetic Miner's process model accurately aligns with the actual revenue processes of the municipality. The score of 0.951 suggests a strong congruence, signifying the model's ability to faithfully represent real-world processes within the context of Hamedan province municipality. This high fitness score underscores the Genetic Miner's proficiency in capturing the nuances of revenue procedures, making it a compelling choice for process analysis and enhancement within this specific domain.

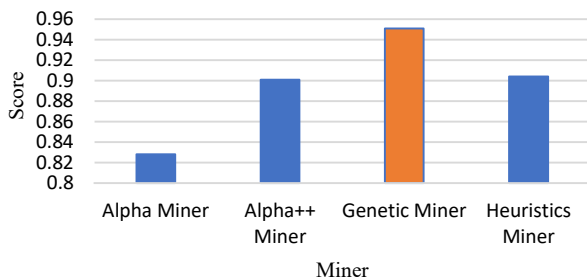


Fig. 7. The best result in fitness

2) Simplicity

As It's shown in Fig. 8., a uniform level of maximum simplicity, consistently rated at 1.0, was attained by all the Miners. This indicates that, regardless of the specific mining method employed, each one excelled in maintaining an optimal level of simplicity in their respective process models.

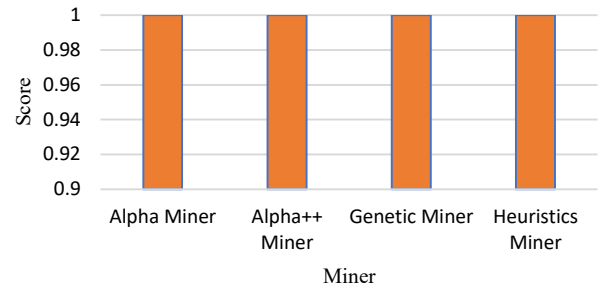


Fig. 8. The best result in simplicity

3) Precision

In reference to Fig. 9., the Genetic Miner exhibited an exceptional level of precision, achieving a perfect score of 1.0. This signifies that the process model generated by the Genetic Miner accurately predicts and adheres to the expected outcomes, reflecting its high level of precision in representing the revenue department processes within the municipality of Hamedan province.

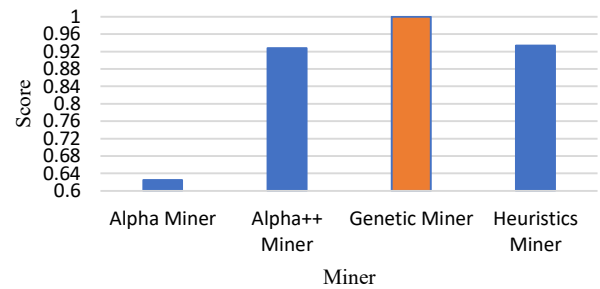


Fig. 9. The best result in precision

4) Generalization

As It's shown in Fig. 10., the Genetic Miner secured the top spot for generalization, achieving an impressive score of 0.978. This outcome underscores the Genetic Miner's remarkable ability to adapt and generalize its process model to diverse scenarios, making it a standout performer in the analysis of revenue department processes within the context of Hamedan province municipality.

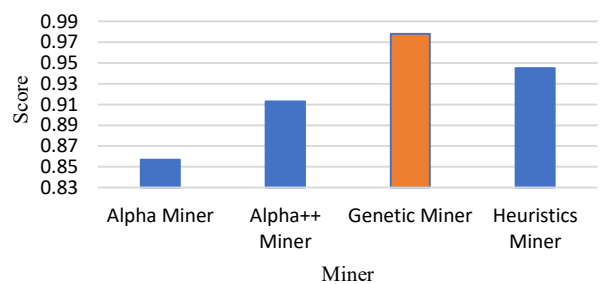


Fig. 10. The best result in generalization

In Fig. 11., a spider chart visually summarizes the attributes of various mining techniques. This chart offers a quick and intuitive comparison of these methods in terms of fitness,

simplicity, precision, and generalization. Each leg of the spider chart represents one of these attributes, and the distance from the center indicates the performance in that dimension. Fitness, simplicity, precision, and generalization are key aspects depicted in the spider chart.

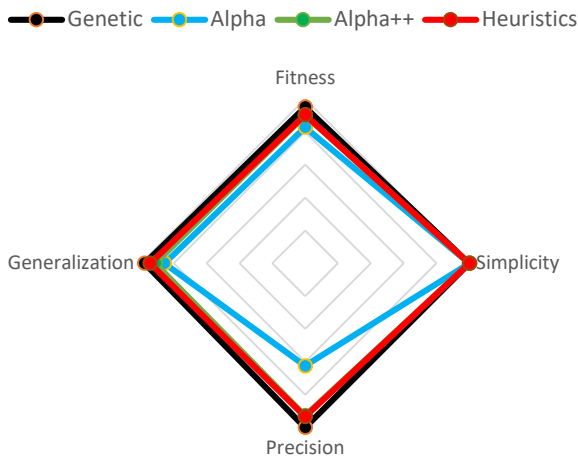


Fig. 11. Fitness, Simplicity, Precision and Generalization (all miners)

B. Choosing the Best Model

As demonstrated in Fig. 12. and Table 1, the standout algorithm in our experiment for analyzing the revenue department processes of Hamedan province municipality was the Genetic Miner, achieving an impressive 98% score in the overall assessment which is an average across key performance metrics including fitness, simplicity, precision, and generalization. Conversely, the Alpha Miner displayed the least favorable performance, with an overall score of 83%. These results underscore the superior effectiveness of the Genetic Miner in this particular context, while highlighting the need for improvements or alternative approaches when considering the Alpha Miner for analysis of the revenue department processes within the municipality.

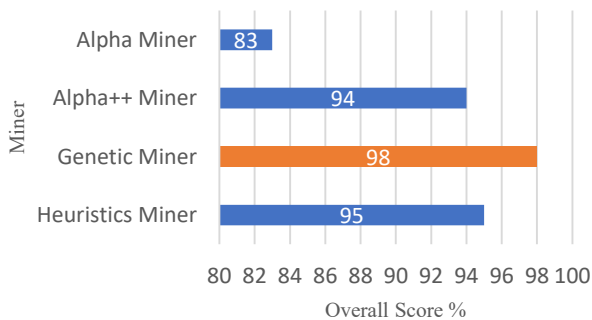


Fig. 12. The best miner (overall score)

TABLE 1. RESULTS

Miner	Fitness	Simplicity	Precision	Generalization	Overall
AM	0.828	1	0.625	0.857	83%
AM++	0.901	1	0.928	0.913	94%
GM	0.951	1	1	0.978	98%
HM	0.904	1	0.934	0.945	95%

V. CONCLUSION AND FUTURE WORK

In the course of this research study, the revenue department processes within the municipality of Hamedan were gathered, documented, and cataloged. Simultaneously, an event log was collected, representing a significant milestone in this investigation. The central objective of this study revolved around the enhancement of the revenue department processes within the municipality of Hamedan. To achieve this, after the methodical consolidation and organization of this event log, it was transformed into an XML file, thereby facilitating its seamless integration into the "ProM 6.12" software environment. Within the confines of this software, the event logs was utilized to generate comprehensive process models, with an exhaustive evaluation undertaken, considering key performance metrics such as fitness, precision, simplicity, and generalization.

The resultant findings stemming from this analytical journey are truly enlightening. The unmistakable standout performer among the miners is the Genetic Miner, which has earned itself an impressive overall score of 98%. This commendable performance is particularly evident in metrics relating to fitness, precision, and generalization, where the Genetic Miner has consistently and remarkably excelled. In stark contrast, the Alpha Miner has demonstrated the least favorable performance, achieving an overall score of 83%, thereby accentuating the necessity for alternative approaches or enhancements when contemplating its utilization in the analysis of revenue department processes within the municipality of Hamedan province. It is of interest to note that, across all miners, the metric of simplicity has yielded uniformly outstanding results, each approach effectively maintaining an optimal level of simplicity in the resulting process models. In light of these compelling results, it is evident that the model generated by the Genetic Miner holds significant promise in the endeavor to enhance the revenue department processes within the municipality of Hamedan among other models.

Looking ahead, the author is firmly committed to not only considering the use of various miners but also to conduct in-depth explorations into more refined models through the application of advanced process mining techniques. This study serves as a robust stepping stone to the continued improvement and optimization of revenue department processes, reinforcing the critical role of process mining in modern municipal management. The pursuit of further research will delve into the nuances of model refinement, performance enhancement, and the development of novel insights, thereby contributing significantly to the advancement of revenue management practices and governance efficiency.

ACKNOWLEDGMENT

I am deeply grateful to Dr. Mansour Esmailpour, whose pivotal role in the development of this paper is beyond measure. His unwavering guidance, unwavering encouragement, and devoted dedication to nurturing my research skills have been invaluable. The support and insightful feedback he provided have unquestionably been the driving force behind the success of this project.

Furthermore, my heartfelt appreciation extends to my family and loved ones, whose unwavering support and unwavering understanding during the arduous research and writing process have been the bedrock of my strength.

In addition to my family, I wish to extend my sincere thanks to Dr. Atieh Jafari for her invaluable guidance, support, and the enlightening feedback she has graciously shared throughout the research process. Her contributions have been instrumental in shaping the outcome of this paper.

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